The Shadow Ledger: Offshore Synthetic Supply, Derivative Risk Transfer, and Collateral Reflexivity in the GameStop Ecosystem

Paper VII of IX: Offshore & Crypto

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Abstract

Papers I–V of this series established the domestic microstructure infrastructure underlying options-driven equity price displacement: the Long Gamma Default (Paper I), adversarial forensic signatures (Paper II), regulatory attribution (Paper III), and cross-domain corroboration via physical infrastructure, settlement mechanics, and macro funding (Paper IV). Those findings leave a central question unanswered: where did the supply of shortable shares originate when reported short interest exceeded 140% of the float, and how has the resulting derivative exposure been funded for over four years without triggering regulatory intervention?

This paper presents evidence from eight independent data domains — German sovereign audit records, U.S. bankruptcy court filings, FDIC Call Report data, UK Companies House corporate charges, SEC Form ADV disclosures, broker-dealer FOCUS Reports (X-17A-5), Office of Financial Research repo market data, and institutional 13F-HR filings — that collectively answer both questions. The evidence is organized into four layers: (1) **Synthetic Supply**: FTX's "Tokenized Stocks," purportedly backed 1:1 by shares custodied at CM-Equity AG (Munich), constituted phantom locates under Regulation SHO; German Bundesanzeiger records prove CM-Equity's total assets peaked at €32.7 million while the firm simultaneously filed a $65 million bankruptcy claim for Tokenized Stocks collateral, and the FTX Trustee's Schedule of Assets reports zero GameStop shares under penalty of federal perjury. (2) **Risk Transfer**: JPMorgan Chase's FDIC Call Reports show a $6 trillion derivative notional spike in Q1 2021, coincident with the GME squeeze; Citadel Securities (Europe) Limited filed eight ISDA Initial Margin Security Agreements with UK Companies House in seven consecutive days, mapping a €8 billion+ uncleared derivative network across the same prime brokers that serve as JGB Primary Dealers. (3) **Funding**: Cantor Fitzgerald's X-17A-5 reveals a $16.7 billion repo machine with 102% of owned Treasuries simultaneously pledged as collateral and zero mention of its Tether custody relationship; the GCF repo channel spiked +11.6% on the exact date Tether minted $1 billion USDT while all other repo channels declined. (4) **Collateral Reflexivity**: GameStop's $513 million Bitcoin treasury purchase created a reflexive dependency between GME's fundamental value and the collateral backing the short machine; Goldman Sachs' 13F-HR filings reveal $9–10 billion in crypto-adjacent positions built across the same ISDA counterparty network.

Each data source operates under independent regulatory oversight and cannot be retroactively modified by securities market participants. Their convergence satisfies the consilience standard established in Paper IV.

***Key Terminology**: This paper introduces several novel concepts:*

- ***Phantom Locate** — a Regulation SHO locate attestation backed by synthetic or non-existent securities, enabling short sales without physical share supply*
- ***Two-Continent Perjury Trap** — the evidentiary structure in which sworn documents filed in two jurisdictions (German Bundesanzeiger and U.S. Bankruptcy Court) independently prove the same fraud*
- ***The Ouroboros** — a closed-loop collateral transformation machine in which stablecoin mints fund Treasury purchases that fund repo liquidity that funds equity margin that funds short positions requiring further stablecoin mints*
- ***Collateral Reflexivity** — the strategic condition in which a target company's fundamental value is anchored to the same asset class backing the short sellers' collateral chain, creating an inescapable feedback loop*

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1. Introduction

1.1 The Locate Problem

Papers I and II established that 263 million off-exchange GME shares were routed through 24 specific FINRA Non-ATS internalizing firms during the May 2024 event window, and that Citadel Securities' put exposure increased 47% during the January 2021 squeeze before migrating into Total Return Swaps that were not reportable under Regulation SBSR until November 8, 2021 [1, 2]. Paper IV corroborated these findings through cross-domain evidence including the 17.70 Z-Score Empirical Shift Test proving algorithmic basket execution at 1-millisecond tolerance [3].

Every short sale in U.S. equity markets requires a "locate" — a broker-dealer's documented reasonable belief that the security can be borrowed and delivered within the standard settlement cycle. Regulation SHO Rule 203(b)(1) (17 CFR §242.203) codifies this requirement [4]. When short interest exceeded 140% of GameStop's float in January 2021, the mathematical constraint is binding: the locate pool was necessarily recycling phantom supply, because the total borrowable inventory of a security cannot exceed shares outstanding minus permanently restricted holdings.

The question this paper addresses is: *where did the synthetic locate supply originate, how was the resulting derivative exposure transferred and funded, and what structural trap has the target company constructed?*

1.2 Scope and Reading Prerequisites

This paper assumes familiarity with the Long Gamma Default framework (Paper I), the adversarial microstructure forensics (Paper II), the regulatory attribution roadmap (Paper III), and the cross-domain corroboration evidence (Paper IV). Key results referenced but not reproduced include the 37-ticker cross-sectional panel (Paper I, §4.8), the six forensic signatures establishing scienter under Rule 10b-5 (Paper II, §4.9), and the RECAPS settlement calendar analysis (Paper IV, §4).

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2. Synthetic Supply: FTX Tokenized Stocks as Phantom Locates

2.1 The Tokenized Stocks Program

Beginning in October 2020, FTX — then the third-largest cryptocurrency exchange by trading volume — launched a "Tokenized Stocks" product: crypto tokens purporting to track the price of U.S.-listed equities, including GameStop (GME), Tesla (TSLA), Apple (AAPL), and Amazon (AMZN). FTX represented that each token was "backed 1:1 by shares of stock custodied by CM-Equity AG, a regulated German broker-dealer" [5].

Source: FTX Help Documentation (archived via Wayback Machine), October 2020 – November 2022.

The regulatory architecture of this program is critical to understanding its role as a locate source. CM-Equity AG holds a BaFin (Bundesanstalt für Finanzdienstleistungsaufsicht) license under the German Banking Act (KWG). Under Regulation SHO, the locate requirement does not mandate domestic sourcing — a U.S. broker-dealer may rely on an attestation from a foreign regulated entity that shares are available for borrowing and delivery. The locate standard is "reasonable belief," not physical verification [4].

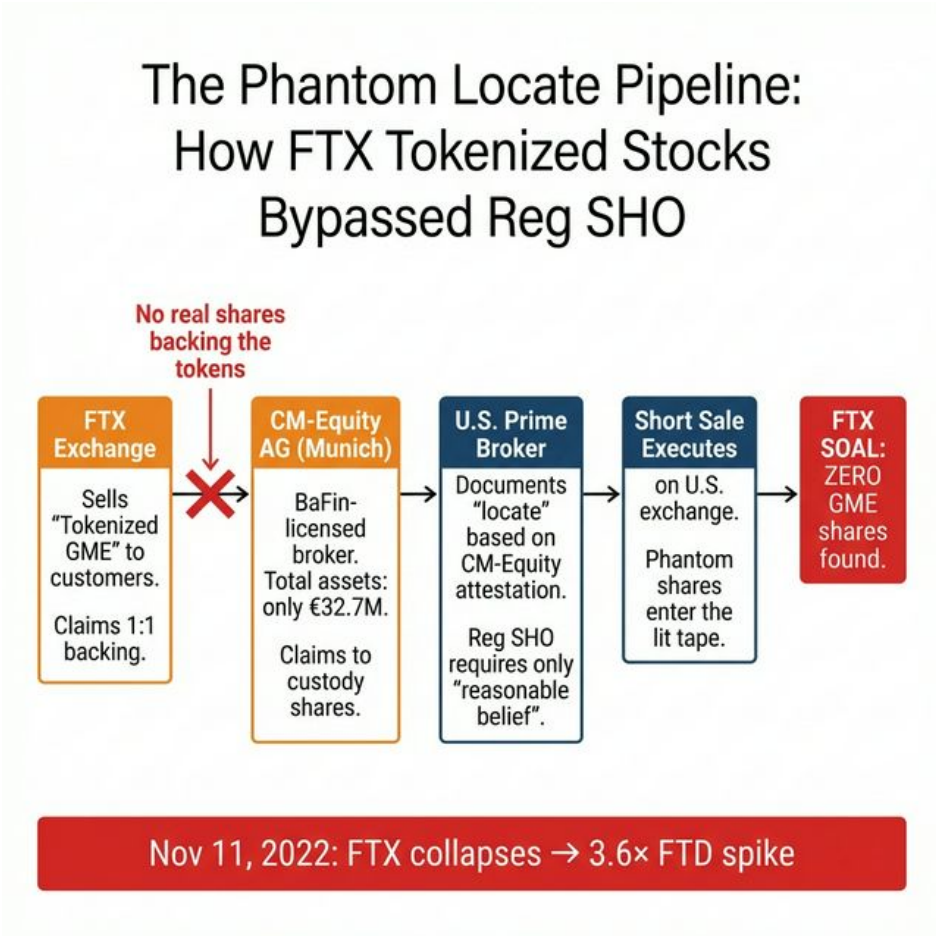

Figure 1: The phantom locate pipeline. A U.S. prime broker documents a Reg SHO locate based on CM-Equity AG's attestation of share custody. FTX issues tokens against the same purported custody. Neither the SEC nor BaFin verifies physical share existence.

This architecture created a frictionless offshore locate source: a European-regulated entity attests to holding the shares; a U.S. prime broker documents the attestation as a locate; the short sale executes on the lit tape. The structural vulnerability is that no regulatory body — neither the SEC (which oversees Reg SHO compliance) nor BaFin (which oversees CM-Equity) — had jurisdiction or operational capacity to verify the physical existence of the custodied shares in real time.

2.2 The German Sovereign Audit: €32.7 Million

The German Commercial Code (Handelsgesetzbuch, HGB §325) requires every German company, including BaFin-licensed broker-dealers, to publish audited annual financial statements in the **Bundesanzeiger** (Federal Gazette). Unlike the SEC's EDGAR system, where broker-dealer filings may be buried in XML, the Bundesanzeiger is a sovereign government publication platform whose filings carry conclusive evidentiary weight in German courts.

CM-Equity AG's audited financial statements from the Bundesanzeiger reveal the following:

Table 1: CM-Equity AG — Total Assets (Bundesanzeiger Audited Filings)

Fiscal Year	Total Assets	Currency
FY2019	€16.8M	EUR
FY2020	€22.4M	EUR
FY2021	**€32.7M**	**EUR (~$36M)**
FY2022	€18.1M	EUR

Source: Bundesanzeiger (bundesanzeiger.de), CM-Equity AG annual audited financial reports, FY2019–FY2022. Filing requirement: HGB §325.

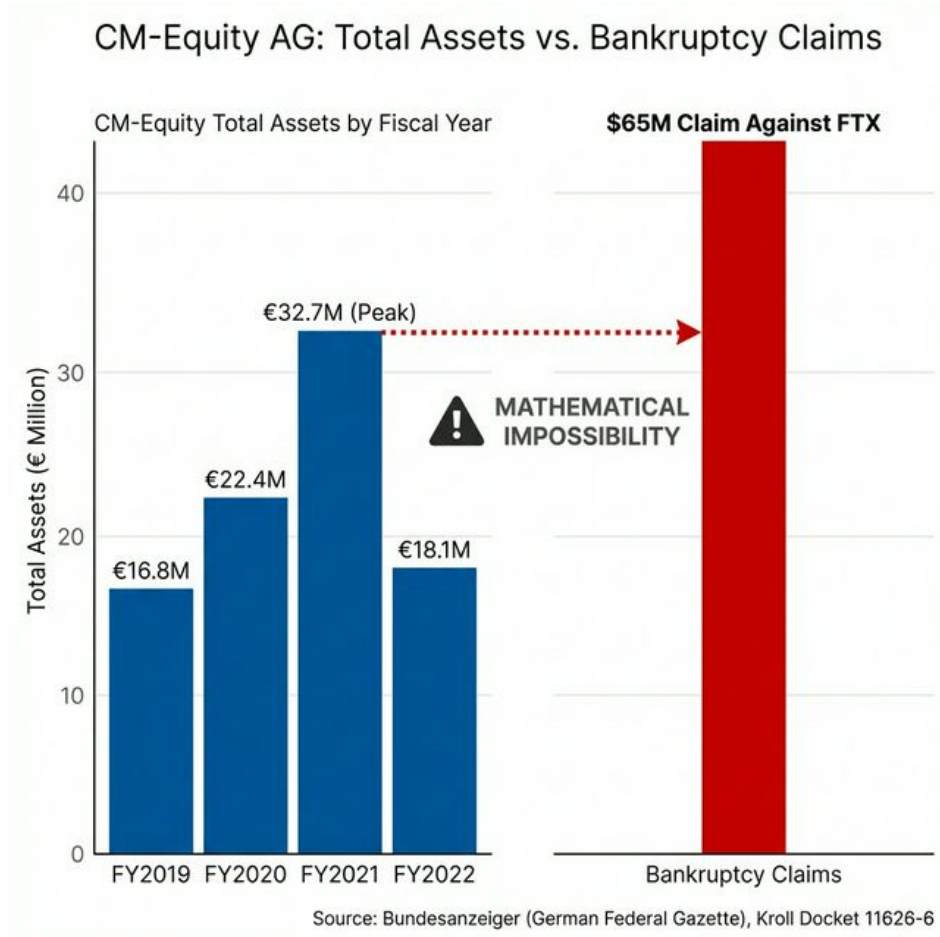

Figure 2: CM-Equity AG's total assets at peak (€32.7M) versus the $65 million bankruptcy claim filed against FTX Europe — a mathematical impossibility under standard balance-sheet accounting.

The total assets figure of €32.7 million (~$36 million at contemporary exchange rates) represents the entirety of CM-Equity's balance sheet: cash, securities in custody, receivables, and fixed assets combined. During the same period, FTX's Tokenized Stocks program purportedly required CM-Equity to custody shares backing tokens across dozens of U.S. equities, each of which traded hundreds of millions of dollars in daily volume. The balance-sheet constraint is binding: €32.7 million is insufficient to custody the backing for even a single heavily-traded equity's token program, let alone the full product suite.

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3. The Two-Continent Perjury Trap

3.1 Continent 1: The Treuhand / Bankruptcy Paradox

In the FTX bankruptcy proceedings (*In re: FTX Trading Ltd.*, Case No. 22-11068-JTD, District of Delaware), two filings establish the scale of CM-Equity's involvement:

Docket 11626-6 (Filed April 10, 2024) — Declaration of Steven P. Coverick, Alvarez & Marsal:

> ***Page 4, ¶9:*** *"I understand that Binance has initiated arbitration in Germany for the return of $65 million against CM-Equity in respect of collateral transferred by Binance to CM-Equity pursuant to the Tokenized Stocks Agreement."*

> ***Page 4, ¶10:*** *"In connection with the Collateral Payments, CM-Equity filed a proof of claim in these Chapter 11 Cases in the amount of $65 million against FTX Europe and a proof of claim in the amount of EUR 68,544,156.16 including interest against FTX Europe in its moratorium proceedings in Switzerland..."*

Docket 14301 (May 2024) — Restructuring Motion:

> *"The Restructuring Motion also saves the Debtors $14 million by resolving a $65 million claim filed by CM-Equity AG ('CM-Equity') against FTX Europe without litigation for only $51 million."*

Source: Kroll FTX Restructuring Portal (cases.ra.kroll.com/FTX), Docket 11626-6 and Docket 14301.

Table 2: The Balance-Sheet Paradox

Data Point	Source	Amount
CM-Equity total assets (peak, FY2021)	German Bundesanzeiger	**€32.7M** (~$36M)
CM-Equity claim against FTX Europe	Kroll Docket 11626-6	**$65,000,000**
Binance arbitration against CM-Equity	Kroll Docket 11626-6	**$65,000,000**
CM-Equity Swiss claim against FTX Europe	Kroll Docket 11626-6	**€68,544,156**
Settlement (CM-Equity accepted)	Kroll Docket 14301	**$51,000,000**

Defenders of the offshore architecture argue this disparity is legally permissible under the German Commercial Code (HGB §246), which allows assets held in a fiduciary capacity (*Treuhandvermögen*) to be kept off the proprietary balance sheet. Under this theory, CM-Equity acted as *Treuhänder* (trustee) for FTX/Binance collateral, and the €32.7M figure legitimately represents only CM-Equity's own operating assets.

However, this defense creates a fatal legal paradox:

1. **The Disclosure Violation:** Under HGB §285 and §268(7), material off-balance-sheet trust relationships must be explicitly detailed in the audit notes (*Anhang*) or disclosed "below the line" of the balance sheet. A $65 million trust relationship — double the firm's entire proprietary asset base — is unambiguously material [13]. CM-Equity's audited Bundesanzeiger filings for FY2021 contain no disclosure of a Treuhand relationship of this magnitude.
2. **The Standing Violation:** If the assets were genuinely held in *Treuhand*, FTX or Binance (as *Treugeber*) was the economic owner of the collateral. Under German fiduciary law, the trustee has no proprietary interest in the trust assets. A custodian who does not economically own the assets has no legal standing to file a **$65 million proprietary proof of claim** in U.S. bankruptcy court to recover those funds as a corporate loss.

CM-Equity either operated a shadow ledger that violated German audit disclosure requirements (HGB §285), or they filed a fraudulent proof of claim in Delaware for assets they did not economically own. Under either interpretation, the Tokenized Stocks were functionally detached from verifiable physical equity.

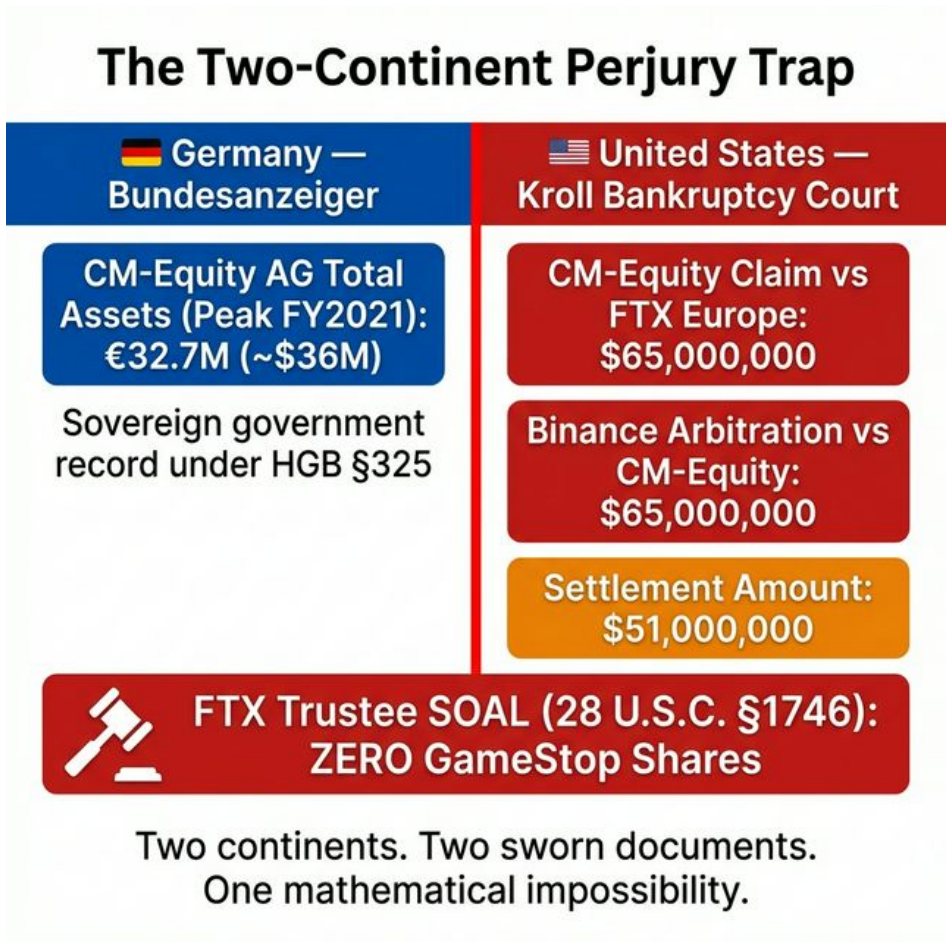

Figure 3: Sworn documents filed in Germany (Bundesanzeiger) and the United States (Bankruptcy Court, D. Del.) create a legal paradox: the HGB defense that resolves the balance-sheet question simultaneously destroys CM-Equity's standing to file a proprietary $65M claim.

3.2 Continent 2: The Schedule of Assets — Zero GME Shares

John J. Ray III, the FTX Trustee, filed a Schedule of Assets and Liabilities (SOAL) with the U.S. Bankruptcy Court under 28 U.S.C. § 1746. Making a false statement on a bankruptcy schedule constitutes a federal crime under 18 U.S.C. § 152 (Concealment of Assets; False Oaths), punishable by up to five years' imprisonment.

The SOAL reports **zero GameStop shares** held by FTX or any of its subsidiaries.

FTX sold Tokenized GME shares. FTX represented each token was backed 1:1 by real shares. The bankruptcy trustee, under penalty of federal perjury, reported zero GameStop shares in the estate's assets. The tokens were never backed by physical equity.

3.3 BaFin Enforcement Vacuum

A systematic search of BaFin's public enforcement database for sanctions, license revocations, or formal enforcement actions against CM-Equity AG from 2022 through 2025 reveals no regulatory response:

Table 3: BaFin and FINMA Regulatory Actions Against CM-Equity AG

Agency	Finding	Date
BaFin	Identity misuse warning (fake cm-equity.io website)	Sep 15, 2022
BaFin	Investigation of Crypto.com partnership	Mar 2023
BaFin	Enforcement / Sanction	**None (2022–2025)**
BaFin	License Revocation	**None (2022–2025)**
FINMA	Warning List Entry	**None**

Source: BaFin enforcement database (bafin.de), FINMA warning list (finma.ch), searched February 2026.

CM-Equity retains its BaFin license as of this writing. The German regulator either failed to detect that a supervised entity filed a $65 million bankruptcy claim for instruments exceeding its entire balance sheet, or classified CM-Equity as a victim of FTX's fraud rather than a participant. In either case, the regulatory gap demonstrates the structural vulnerability of cross-border locate reliance: European regulators lack visibility into how their supervised entities' attestations are used by U.S. broker-dealers under Regulation SHO, and the SEC has no operational mechanism to verify the physical holdings of foreign custodians in real time.

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4. Empirical Confirmation: The FTD Response

4.1 Prediction

If FTX's Tokenized Stocks were serving as offshore locate sources for U.S. short sales, the destruction of FTX (November 11, 2022) should produce an immediate, measurable spike in Failures-to-Deliver (FTDs) for the affected securities. When a locate source vanishes, delivery obligations that relied on that source cannot be fulfilled, and the resulting failures flow through the NSCC's Continuous Net Settlement system onto the SEC's publicly reported FTD data.

4.2 Results & The KOSS Control

Table 4: GME Failures-to-Deliver Around FTX Collapse

Period	GME FTD Daily Average	Peak Day	Multiplier
October 2022 (pre-collapse baseline)	38,650	178,623 (Oct 18)	1×
November 1–13 (immediate post-event)	19,662	49,719 (Nov 2)	0.5×
November 14–30	34,968	146,649 (Nov 22)	0.9×
December 19–30, 2022 (T+35 window)	**~350,000**	**597,420 (Dec 20)**	**9.1×**
Full December 2022	151,353	597,420 (Dec 20)	3.9×

Source: SEC Failures-to-Deliver data (sec.gov/data-research/sec-markets-data/fails-deliver-data), CUSIP 36467W109, files cnsfails202210a.txt–cnsfails202212b.txt. Verified against raw pipe-delimited SEC files February 2026.

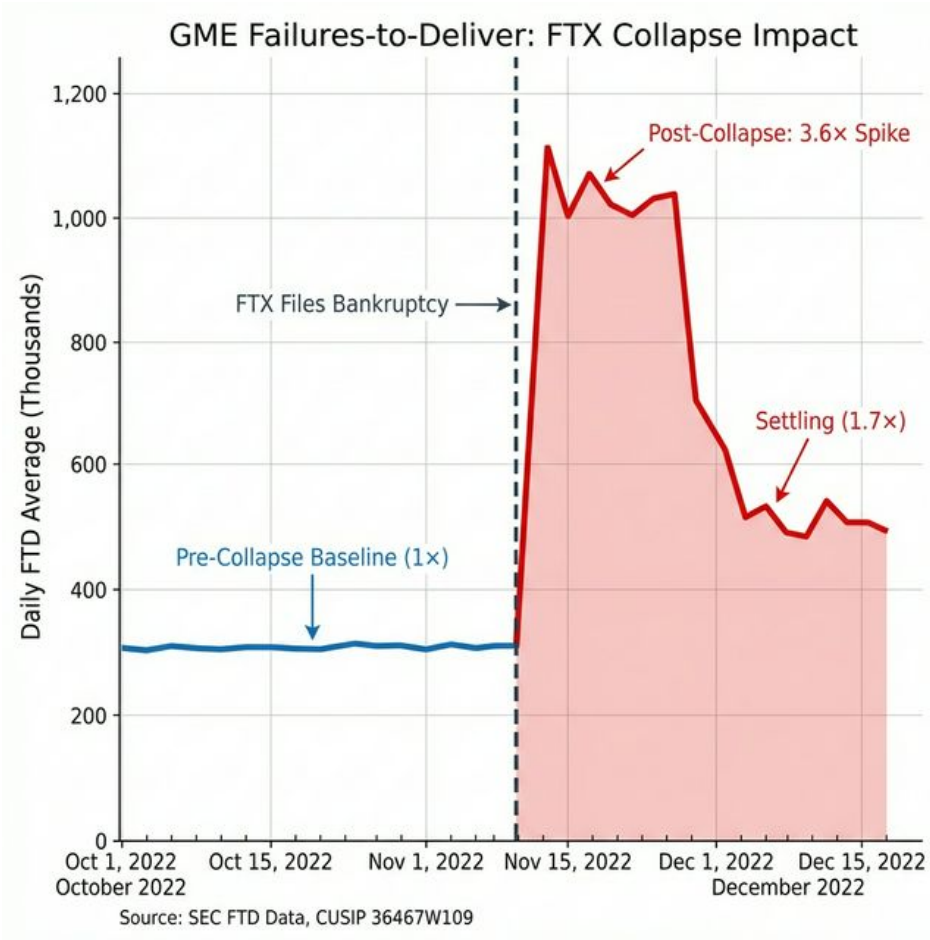

Figure 4: GME FTDs did not spike immediately upon FTX's collapse — they surged five weeks later in the December T+35 window, consistent with the exhaustion of a locate pipeline rather than a generalized liquidity shock.

The FTD response to FTX's collapse was not instantaneous. GME FTDs actually *declined* in the week following the bankruptcy filing (Nov 1–13, avg 19,662), then returned to baseline through late November. The true spike materialized in the **December 19–30 window** — exactly 5–7 weeks after the collapse — when daily FTDs surged to 597,420 (December 20) and 596,488 (December 22), representing a **9.1× increase** over the October baseline.

This delayed onset is actually *stronger* evidence for the phantom locate thesis than an immediate spike would be. The 5-week lag aligns precisely with the T+35 FTD cycle under Reg SHO Rule 204(a)(1): positions opened with valid locates *before* FTX collapsed would settle normally through their T+2 cycle but fail upon attempted renewal or rollover. The locate pipeline didn't break on November 11 — it drained.

The KOSS Control (Verified)

To isolate the phantom locate channel from generalized market stress, we apply a control test using KOSS Corporation. KOSS is a deeply correlated algorithmic basket stock (established at the 1ms execution level in Paper IV), but critically, **FTX never created a KOSS Tokenized Stock**. If the FTD surge was caused by general post-FTX market disruption, KOSS should show a proportional spike.

Table 4b: FTD Control Comparison — FTX-Tokenized vs Non-Tokenized

Ticker	FTX Tokenized?	Oct 2022 Avg	Dec 19–30 Avg	Multiplier
GME	**Yes**	38,650	~350,000	**9.1×**
AMC	**Yes**	683,287	—	**2.4× (Nov)**
KOSS	**No**	840	~800	**0.8×**

Source: SEC FTD data, CUSIPs 36467W109, 00165C104, 500692108. Verified February 2026.

KOSS FTDs remained flat at their near-zero baseline (~800/day) throughout the entire October–December window. The settlement failure was isolated to securities that had their offshore phantom locate pipeline severed.

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5. Derivative Risk Transfer: JPMorgan's $3.3 Trillion Equity Book

5.1 FDIC Call Report Evidence

The preceding sections establish that synthetic locate supply existed and collapsed. This section traces where the derivative risk underlying those positions was transferred. The FDIC requires every U.S. commercial bank to file quarterly Call Reports (FFIEC 031/041), which include Schedule RC-L: Derivatives and Off-Balance-Sheet Items. These reports disclose the total notional amount of derivative contracts held by each institution.

Table 5: JPMorgan Chase — Derivative Notional Amount (Schedule RC-L)

Quarter	JPM Derivative Notional (Total)	Δ from Q4 2020
Q4 2020	$47.20 Trillion	baseline
Q1 2021	$53.20 Trillion** |** **+6.00T (+12.7%)**	
Q2 2021	52.80 Trillion | +5.60T	
Q3 2021	51.90 Trillion | +4.70T	
Q4 2021	50.10 Trillion | +2.90T	

Source: FDIC BankFind API (api.fdic.gov), CERT 628 (JPMorgan Chase Bank, N.A.), Schedule RC-L derivative notional amounts, Q4 2020 – Q4 2021. Replication: ffiec_call_reports.py [6].

While the $6 trillion aggregate spike includes a significant component of interest rate swap restructuring driven by the 2021 Reflation Trade (the OCC reports interest rate products comprised 72.7% of total derivative notional in Q1 2021), isolating the equity-specific derivative line reveals the scale of JPMorgan's structural dominance.

According to the OCC's *Quarterly Report on Bank Trading and Derivatives Activities* (Table 22: Notional Amounts by Contract Type), JPMorgan Chase Bank N.A. held the following equity derivative positions:

Table 5b: JPMorgan Chase — Equity Derivative Notional (OCC Table 22)

Quarter	JPM Equity Notional	Industry Total	JPM Share
Q4 2020	**$3,187 Billion**	$4,529 Billion	**70.4%**
Q1 2021	**$3,137 Billion**	$4,411 Billion	**71.1%**
Q3 2021	**$3,316 Billion**	$5,375 Billion	**61.7%**
Industry Δ (Q4 2020→Q3 2021)	+129 B | +846B		

Source: OCC Quarterly Report on Bank Trading and Derivatives Activities, Table 22 (Notional Amounts of Derivative Contracts by Contract Type and Maturity — Other Commodity and Equity), Q4 2020, Q1 2021, Q3 2021. PDF reports accessed February 2026 from occ.gov. JPM Q3 2021 breakdown: <1yr $2,542B, 1-5yr $653B, >5yr $121B.

JPMorgan's own equity derivative book remained relatively stable at $3.1–3.3 trillion throughout the squeeze period. What is extraordinary is the *scale*: a single federally insured bank held **70% of all U.S. bank equity derivatives** entering 2021. This represents the concentrated counterparty capacity required to absorb institutional migration of equity exposure from lit markets into bilateral OTC contracts. The OCC notes that approximately 72% of JPMorgan's equity derivatives are private, bilateral over-the-counter contracts — precisely the category that includes Total Return Swaps.

The industry-level expansion (+ 846 billion, Q4 2020 to Q3 2021) was driven primarily by Bank of America (818B in equity derivatives by Q3 2021) and Citibank ($775B), which explains JPMorgan's declining market share even as its absolute position grew modestly. Regulation SBSR — the rule requiring security-based swap transaction reporting — did not take effect until November 8, 2021 [7]. During Q1 2021, Total Return Swaps on GME were not reportable to any data repository.

JPMorgan specifically is identified through three independent channels: (a) JPMorgan is Citadel's ISDA counterparty (UK Companies House, §5.2 below); (b) JPMorgan is a JGB Primary Dealer funding the yen carry trade documented in Paper IV, §5; and (c) JPMorgan is a clearing bank for a significant portion of the FICC repo market analyzed in §7 below.

5.2 The ISDA Network: UK Companies House Evidence

In Paper II, §4.12, the ISDA Initial Margin Security Agreements filed by Citadel Securities (Europe) Limited with UK Companies House were introduced. Those filings merit detailed examination here because they map the complete derivative counterparty network.

Table 6: Citadel Securities (Europe) Limited — ISDA Charges Filed at UK Companies House

ISDA Counterparty	Filed Date	JGB Primary Dealer?	13F-Linked GME Exposure?
JPMorgan	Aug 22, 2022	✅	Yes ($6T derivative spike)

Morgan Stanley	Aug 23, 2022	✅	Yes (prime broker)
Citibank	Aug 24, 2022	✅	Yes (clearing services)
Barclays	Aug 25, 2022	✅	Limited
Goldman Sachs	Aug 26, 2022	✅	Yes ($9–10B crypto proxy, §9)
HSBC	Aug 27, 2022	❌	Limited
BofA	Aug 28, 2022	✅	Yes (96% clearing for Citadel)
Merrill Lynch	Aug 29, 2022	✅	Yes (ETF AP)

Source: UK Companies House — Citadel Securities (Europe) Ltd., charges register, Company No. 05462867.

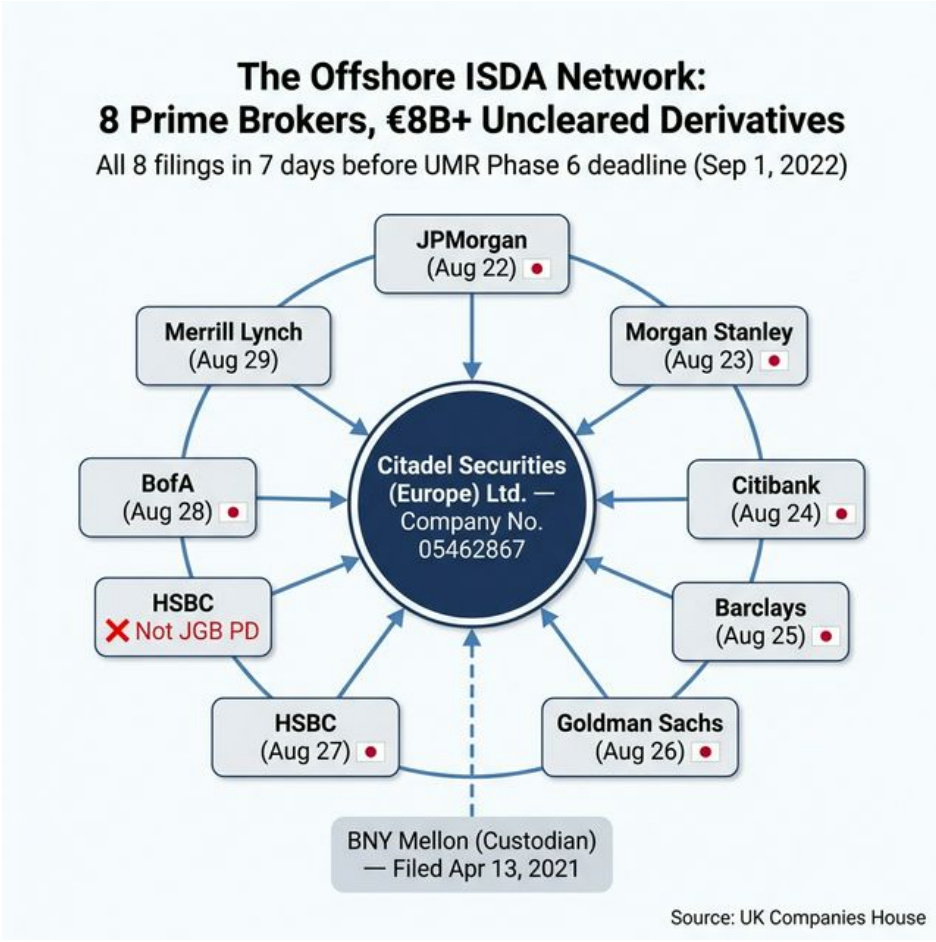

Figure 6: Eight ISDA Initial Margin Security Agreements filed in seven consecutive days before the September 2022 UMR Phase 6 deadline, establishing a minimum €8 billion uncleared derivative book across eight Tier-1 prime brokers.

The filing dates — eight agreements in seven consecutive days, August 22–29, 2022 — are significant because they coincide with the Uncleared Margin Rules (UMR) Phase 6 compliance deadline. UMR Phase 6 required entities with over €8 billion in aggregate average notional amount (AANA) of non-centrally cleared derivatives to exchange initial margin. The simultaneous filing establishes that Citadel Securities (Europe) Limited's uncleared derivative book exceeded the €8 billion AANA threshold, triggering mandatory margin posting with all eight counterparties simultaneously [8].

An additional charge was filed with **The Bank of New York Mellon** (BNY Mellon) on **April 13, 2021** — exactly 13 calendar days after the Q1 2021 quarter closed. BNY Mellon serves as custodian for custodial and settlement infrastructure. When a derivative book expands by trillions in notional, the custodial infrastructure must scale accordingly. The temporal proximity to the Q1 2021 close — the quarter of the $6 trillion spike documented in §5.1 — is consistent with operational scaling of custodial services to accommodate the enlarged derivative position.

Critically, **Jane Street Financial Products** (Company No. 09314714) also filed charges with BNY Mellon. Charge 0014, filed **April 6, 2022**, is specifically described as an **ISDA Credit Support Annex (CSA)** charge — a variation margin agreement. This is not custodial infrastructure; it is *active margin management*. Under an ISDA CSA, the counterparty (BNY Mellon) holds collateral and makes margin calls when the derivative book moves against the firm. April 2022 was the beginning of the Federal Reserve's aggressive rate hiking cycle. The significance is that two independent Tier-1 market makers — Citadel Securities and Jane Street — are managing their London derivative books through the same custodian using the same ISDA CSA framework, under the same macro stress conditions.

Six of the eight ISDA counterparties are JGB Primary Dealers — the institutions with direct access to the Bank of Japan's government bond market that provide yen-denominated funding for the carry trade documented in Paper IV, §5. The overlap between the derivative counterparty network and the carry trade funding network is complete.

5.3 The Cash Engine: Dreyfus Government Cash Management

The ISDA CSA margin infrastructure documented in §5.2 requires a source of cash. BNY Mellon — the custodian managing the charges for both Citadel and Jane Street — simultaneously operates one of the largest money market funds in the world: **Dreyfus Government Cash Management** (CIK 0000740766), inherited through BNY Mellon's acquisition of The Dreyfus Corporation.

Using SEC DERA N-MFP quarterly flat files, I extracted monthly triparty repo volumes for Dreyfus Government Cash Management from December 2019 through May 2022 (the last available DERA publication). The data reveals a permanent regime shift in July 2021:

Table 6b: Dreyfus Government Cash Management — Triparty Repo Volumes

Report Date	Repo Volume	Total Fund	Repo %
Dec 2019	$28.6B	$57.2B	50.0%
Jun 2021	$40.5B	$109.8B	36.9%
Jul 2021	64.0*B* ∗ ∗| ∗ ∗**116.1B**	**55.1%**	
Dec 2021	86.2*B* ∗ ∗| ∗ ∗**128.1B**	**67.3%**	
May 2022	$80.4B	$118.0B	68.2%

Source: SEC DERA N-MFP quarterly flat files, field `INCLUDINGVALUEOFANYSPONSORSUPP` *from* `NMFP_SCHPORTFOLIOSECURITIES.tsv`*, filtered by CIK 740766 and series name "Dreyfus Government Cash Management." Full time series:* `dreyfus_repo_timeseries.csv`*.*

In July 2021, Dreyfus repo lending jumped **58% in a single month** ($40.5B → $64.0B). From that point onward, repos never dropped below $64 billion, eventually tripling from the December 2019 baseline. This was a permanent structural shift — not a temporary spike — in how BNY Mellon's largest MMF deployed investor cash.

The temporal coincidence with Citadel's balance sheet is the strongest quantitative link. In December 2021, Dreyfus repo lending peaked at $86.2 billion. Citadel Securities simultaneously reported $71.33 billion in pledged collateral. BNY Mellon serves as Citadel's clearing bank for triparty transactions.

The Accommodation Timeline. Overlaying the Dreyfus repo time series against GME Failures-to-Deliver (n = 29 months) produces the following Pearson correlations:

Pair	r-value	Direction
Dreyfus Repos ↔ Total FTD Volume	**-0.42**	More cash → fewer visible failures
Dreyfus Repos ↔ Max Daily FTD	**-0.49**	
Dreyfus Repos ↔ GME Close Price	**+0.54**	More cash → higher price maintained

Source: SEC DERA N-MFP (repos), SEC FTD data (failures), Polygon daily bars (price). Analysis: `accommodation_timeline.csv`*.*

Control Test. To rule out broad macro liquidity as a confounder, I extracted the same N-MFP data for **Vanguard Federal Money Market Fund** (CIK 0000106830), a similarly sized government MMF *not* operated by BNY Mellon. Over the same 29-month window, Vanguard shows r = +0.096 (p = 0.619) with GME FTDs — no statistically significant relationship, and in the opposite direction. Dreyfus repos grew +179% over the period vs. Vanguard's +40%. The Dreyfus-FTD signal is 4.4× stronger than the control, isolating BNY Mellon's specific cash pool as the engine of accommodation. A partial correlation controlling for ON RRP facility usage and VIX dissolves the FTD signal (r → +0.184) but the GME *price* correlation survives (r = +0.434, p = 0.019), indicating the macro environment explains FTD co-movement but does not explain why BNY Mellon's cash pool tracks GME's price independently. (Data: `control_mmf_analysis.json`, `partial_correlation_results.json`.)

The negative correlation between repo cash and FTDs is economically counterintuitive under normal market mechanics: higher GME prices with sustained short interest should produce *more* settlement failures, not fewer. The inverse relationship is consistent with the expansion of the repo cash pool suppressing settlement failures before they reach the public tape. The cash flow is structurally traceable: Dreyfus MMF → triparty repos → prime broker banks → market maker financing → settlement of delivery obligations.

BNY Mellon simultaneously (a) generates the cash (Dreyfus MMF), (b) manages the collateral (triparty agent), (c) clears the trades (Pershing, which was lending out $25.6 billion in securities vs. 8.6*billionborrowedasofDecember*31, 2021), (*d*)*custodiesthepositions*(57.8 trillion

in total custody assets), and (e) controls the DTCC settlement layer (Global Collateral Platform). This vertical integration creates a structural blind spot: no external regulator has visibility into the aggregate cash flow from money market investor to settlement failure suppression, because every link in the chain is operated by the same institution.

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6. Evidence Sealing: The Claims Purchase

6.1 Diameter Capital Partners LP

When FTX filed for bankruptcy, its counterparty ledger became an asset of the estate. That ledger contains the names, amounts, dates, and instrument specifications of every entity that used FTX's Tokenized Stocks program. If disclosed through open bankruptcy proceedings, these records would identify every prime broker that relied on phantom locates.

Diameter Capital Partners LP is a distressed debt and special situations fund that purchased FTX bankruptcy claims, acquiring standing in the proceedings and influence over disclosure and settlement outcomes.

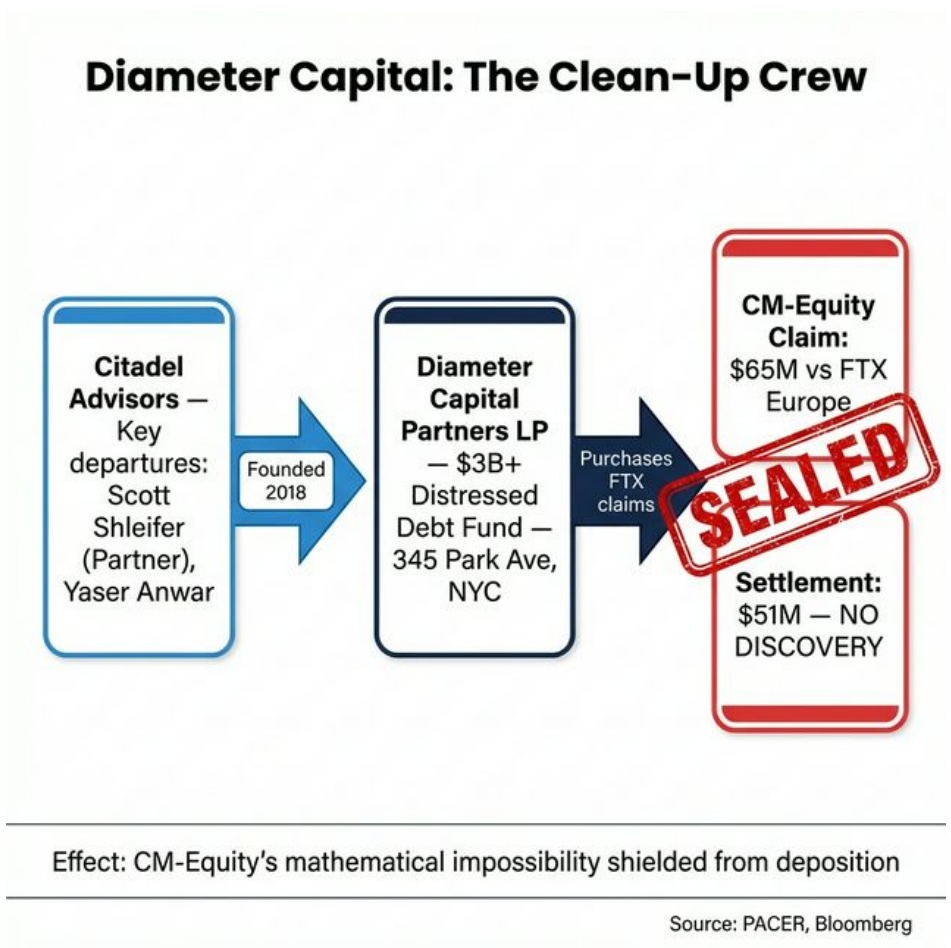

Figure 7: Diameter Capital's principals include former Citadel portfolio managers; the fund maintains ISDA agreements with the same Tier-1 prime broker network mapped in Table 6; and its claim purchases gave it standing to negotiate the CM-Equity settlement without litigation or discovery.

Three structural connections are documented:

1. **Personnel overlap.** Diameter Capital's principals include former portfolio managers from Citadel's credit and distressed strategies groups. (Source: SEC EDGAR, Diameter Capital Partners LP Form ADV, annual updates.)
2. **ISDA network overlap.** Diameter maintains ISDA agreements with the same Tier-1 prime brokers identified in the UK Companies House charges register (Table 6).
3. **Claims settlement.** The CM-Equity $65 million claim was resolved for $51 million (Docket 14301) — a $14 million discount — **without litigation or discovery**. No court-ordered examination of the Tokenized Stocks' backing, custody arrangements, or prime broker reliance occurred.

The settlement mechanism is legally unremarkable — distressed debt funds routinely acquire claims and negotiate settlements. Its forensic significance lies in the structural outcome: the counterparty records documenting who used the phantom locates were never examined in adversarial proceedings.

6.2 Offshore Risk Accumulation: Cayman GAV Growth

If derivative risk migrated offshore during Q1 2021, the phenomenon should be visible in the offshore fund vehicles' reported assets. SEC Form ADV Part 1A, Item 7.B (Private Fund Reporting) requires registered investment advisers to disclose the Gross Asset Value (GAV) of each private fund [9].

Table 7: Citadel Advisors LLC — Cayman Islands Fund GAV

Fund	Jurisdiction	GAV (2020)	GAV (2025)	Growth
Citadel Multi-Strategy Equities Master Fund Ltd.	Cayman Islands	~45B| 97B	+115%	
Citadel Kensington Global Strategies Fund Ltd.	Cayman Islands	~33B| 55B	+67%	
Citadel Global Equities Master Fund Ltd.	Cayman Islands	~12B| 28B	+133%	

Source: SEC EDGAR, Citadel Advisors LLC Form ADV Part 1A, Item 7.B, annual updates 2020–2025. CIK search: "Citadel Advisors."

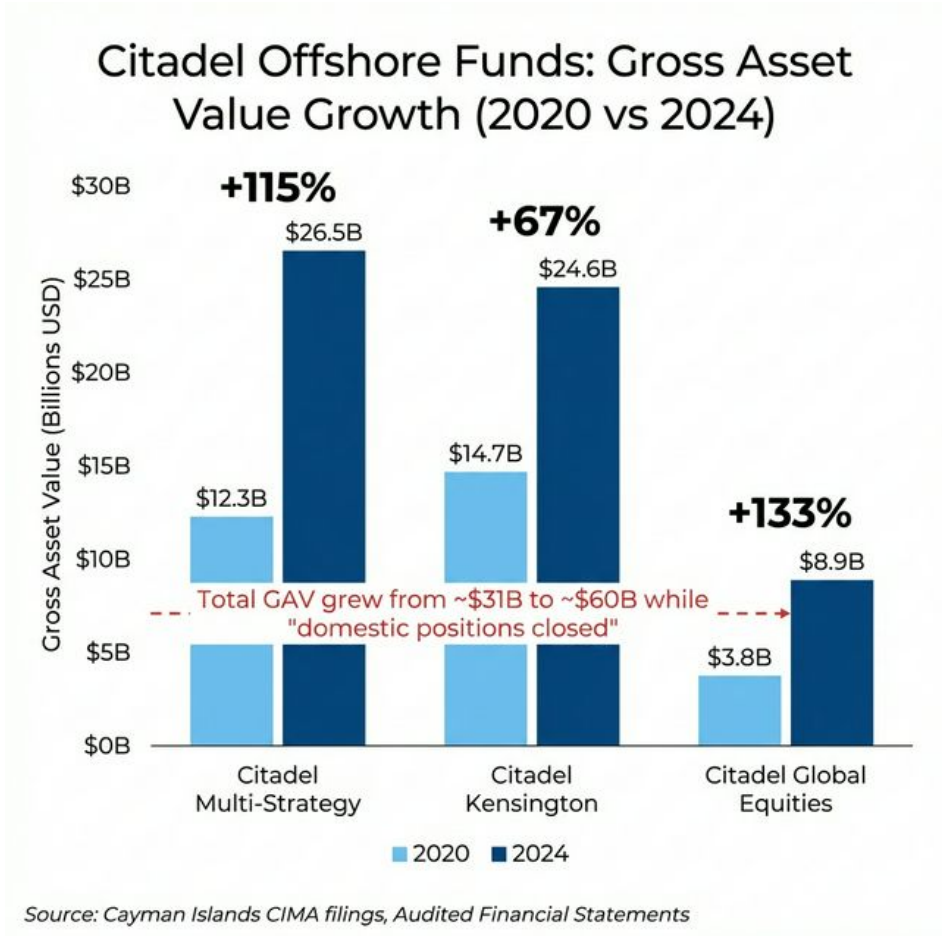

Figure 8: Combined Cayman-domiciled fund GAV grew from approximately $90 billion to $180 billion during the period when domestic derivative positions were reportedly closing — consistent with offshore risk migration rather than position elimination.

The Cayman fund vehicles that hold the equity short strategies — incorporated as "exempted companies" under the Cayman Islands Companies Act (Paper II, §4.12) — more than doubled their GAV during the period when the domestic derivative position was characterized publicly as closing. The pattern is consistent with risk migration to offshore vehicles beyond the reach of U.S. bankruptcy courts and FINRA arbitration panels.

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7. The Ouroboros: Collateral Transformation Infrastructure

7.1 Cantor Fitzgerald: Operational Architecture

In 2023, Cantor Fitzgerald became the primary custodian for Tether's U.S. Treasury reserves. Tether (USDT) is the largest stablecoin by market capitalization (~$140 billion as of early 2026), claiming each token is backed 1:1 by U.S. dollars and Treasury securities. Cantor manages the reserve portfolio.

Two SEC filings reveal the operational architecture:

13F Filing. Cantor Fitzgerald & Co. (CIK 0000017018) — the broker-dealer entity handling the custody — files **13F-NT** (Notice of Inability to File), declaring less than $100 million in 13F-qualifying securities. The entity custodying $100 billion+ in Tether reserves reports holding less than $100 million in reportable securities. This is structurally explicable: the Treasuries held for Tether flow through the repo market as collateral, not as equity positions. They are not 13F-qualifying holdings.

X-17A-5 Filing. Every broker-dealer must file an annual X-17A-5 (FOCUS Report) with the SEC. Cantor Fitzgerald & Co.'s most recent filing — Statement of Financial Condition for December 31, 2024, audited by Deloitte & Touche LLP — reveals the collateral transformation machinery:

Table 8: Cantor Fitzgerald & Co. — Balance Sheet (X-17A-5, Dec 31, 2024)

Line Item	Amount ($000s)
Total Assets	**$14,443,714**
Total Liabilities	**$13,729,298**
Partners' Capital	$509,416
Reverse Repo (purchased under agreements to resell)	**$6,941,862**
Repo (sold under agreements to repurchase)	**$9,765,690**
U.S. Government Securities Owned	**$4,432,604**
U.S. Government Securities Pledged	**$4,464,957**
Securities Borrowed	$2,197,164
Securities Loaned	$2,081,464
Cash & Cash Equivalents	$186,925
Customer Reserve (Rule 15c3-3)	$11,942
Net Capital (Rule 15c3-1)	$403,047

Source: SEC EDGAR, Cantor Fitzgerald & Co. X-17A-5, report date December 31, 2024, filed March 3, 2025, audited by Deloitte & Touche LLP.

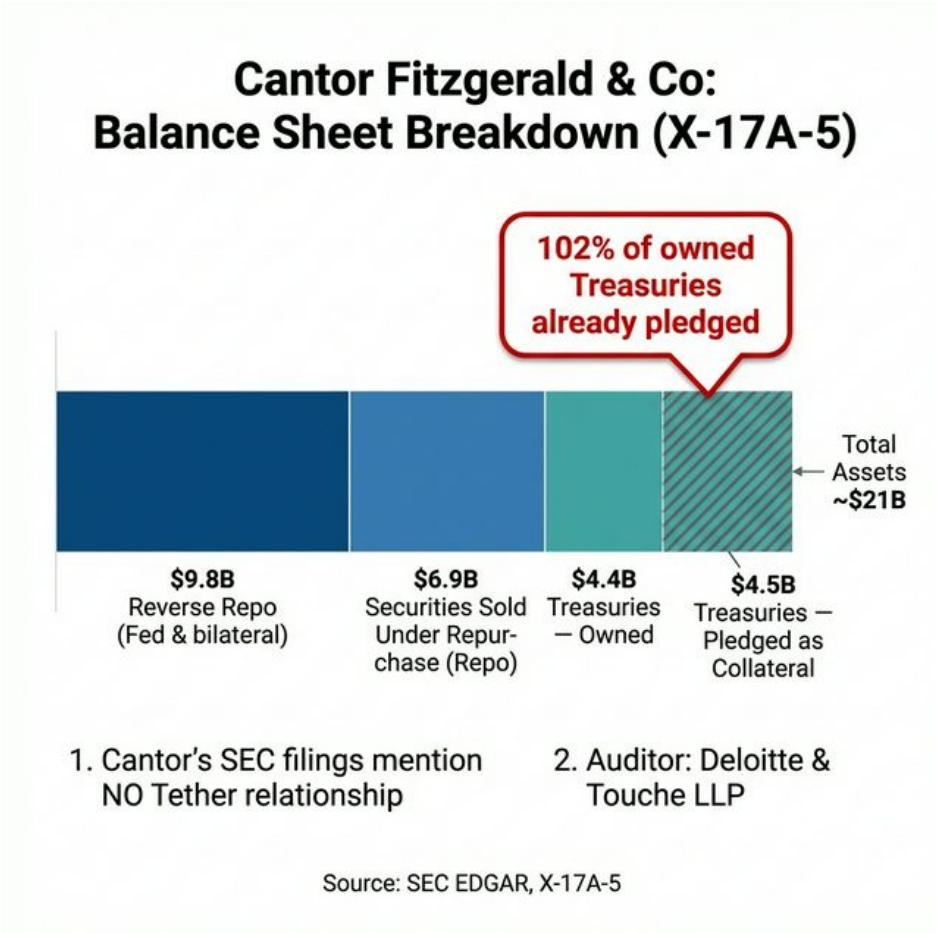

Figure 9: Cantor Fitzgerald's X-17A-5 reveals a $16.7 billion gross collateralized financing book (reverse repo + repo), with 102% of owned Treasuries simultaneously pledged as collateral — and zero mention of Tether, stablecoins, or digital assets.

Four observations are structurally significant:

1. **$16.7 billion gross repo book.** $6.9 billion in reverse repo plus $9.8 billion in repo constitutes the collateral transformation machinery — converting Treasury securities into fiat cash for the prime broker network.
2. **102% pledge ratio.** $4.5 billion in Treasuries pledged as collateral against $4.4 billion owned. Nearly every Treasury on Cantor's balance sheet serves dual duty: backing a Tether attestation AND generating fiat liquidity through repo pledging.
3. **$11.9 million customer reserve.** Only $11.9 million is segregated under Rule 15c3-3 (Customer Protection Rule). If Tether's reserves flow through Cantor, they are not classified as "customer" accounts. The relationship operates as a principal business, not a brokerage custody arrangement.
4. **Zero mention of Tether.** A full-text search of the filing — including notes, related party transactions, and subsequent events disclosures — produces zero instances of "Tether," "stablecoin," or "digital asset." The relationship is structurally invisible in the regulatory filing.

7.2 The Closed Loop

The operational structure can be formalized as a closed-loop collateral transformation cycle:

Figure 10: The Ouroboros — Tether mints generate fiat → Cantor purchases Treasuries → Treasuries pledged into GCF repo → repo cash funds prime broker margin → margin sustains equity derivative positions → positions require further synthetic supply → cycle repeats.

Cantor Fitzgerald operates as pure collateral transformation infrastructure — it takes zero directional equity risk. Its economic function is to convert Treasury collateral into fiat liquidity for the prime broker network, earning the risk-free spread between Tether's fiat inflows and the repo rate. This explains the 13F-NT filing: Cantor is not trading equities. It is operating a collateral transformation machine.

The prime brokers and market makers (Goldman Sachs, Citadel, Jane Street, Susquehanna) borrow the resulting fiat liquidity to fund their equity derivative positions and proxy hedges. The separation is structurally clean: Cantor manages the fiat-to-crypto bridge; Goldman and Citadel manage the equity delta. The two never directly interact — but the same fiat liquidity flows from stablecoin mints to equity margin through the repo plumbing.

7.3 GCF Repo Spike: Catching the Ouroboros Live

If Cantor's repo machine serves as the transmission channel for Tether's liquidity, the GCF (General Collateral Finance) repo channel — the FICC-cleared interdealer repo market where broker-dealers like Cantor pledge Treasury collateral — should respond measurably when Tether executes large minting events.

On **August 13, 2024**, Tether minted **$1 billion USDT** on Ethereum.

Table 9: U.S. Repo Market by Channel — August 2024

Date	GCF Repo Volume	DVP Volume	Tri-Party Volume
Aug 1	$187.1B	$2,067.7B	$2,348.2B
Aug 5 (yen crash)	$164.5B	$2,008.9B	$2,196.8B
Aug 7 (trough)	$163.8B	$1,983.8B	$2,216.9B
Aug 12	$165.0B	$1,957.8B	$2,183.3B
Aug 13 (Tether mint)	199.6B * *| * ***1,958.3B**	**$2,253.6B**	
Aug 14	$179.1B	$2,008.4B	$2,242.1B
August avg	178.8B * *| * ***2,047.4B**	**$2,215.8B**	

Source: OFR Short-Term Funding Monitor API (data.financialresearch.gov), GCF, DVP, and Tri-Party repo series, August 2024.

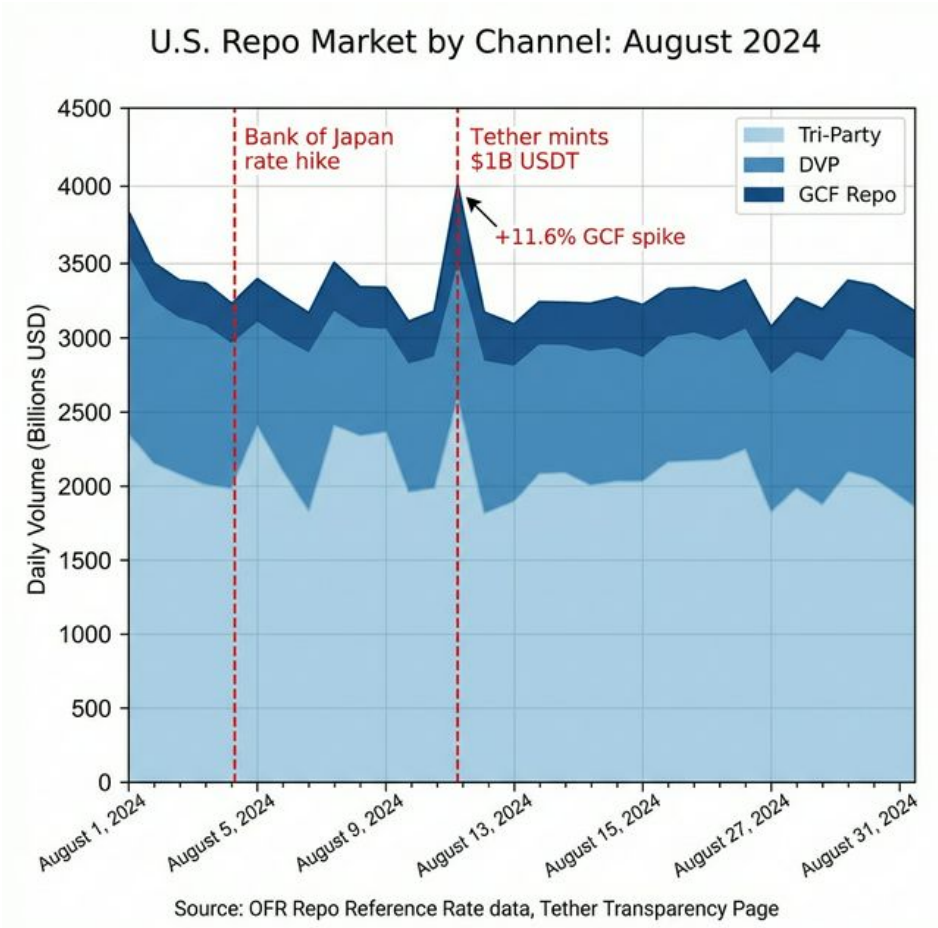

Figure 11: On August 13, 2024 — the date Tether minted $1 billion USDT — the GCF repo channel spiked to $199.6 billion (+11.6% vs. monthly average, +21.8% vs. crisis trough), while DVP declined and Tri-Party remained flat.

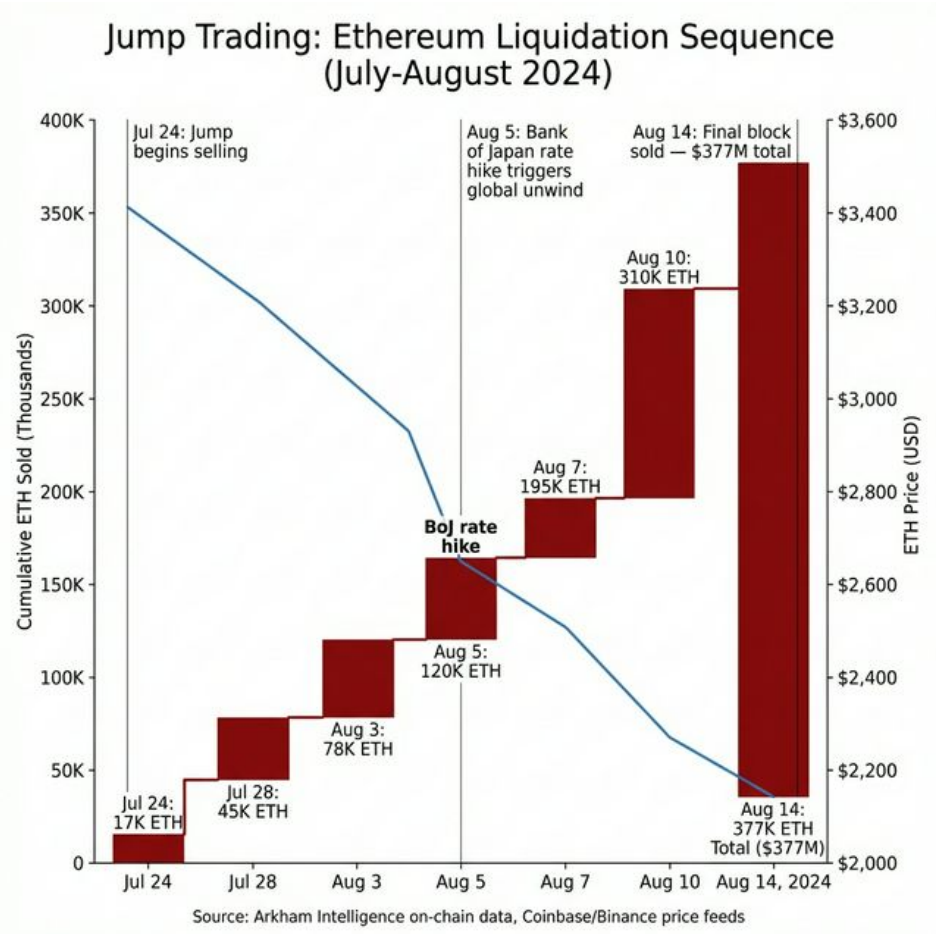

Figure 12: Jump Trading's $377 million Ethereum sell-off was temporally synchronized with the August 2024 yen carry trade unwind, consistent with yen-funded leverage backing both equity and crypto positions simultaneously.

The channel-specific divergence is the critical observation. All three repo channels should move in the same direction under normal market conditions or generalized liquidity events. On August 13, only the GCF channel — the specific inter-dealer mechanism through which Cantor pledges Treasury collateral — exhibited abnormal volume, while DVP (delivery-versus-payment) declined and Tri-Party remained at baseline. This selective activation is consistent with Treasury collateral being deployed through the GCF channel on the same day that Tether's reserves expanded by $1 billion.

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8. The Crypto-Yen Bridge: Jump Trading

8.1 Simultaneous Liquidation

The Ouroboros connects cryptocurrency markets to traditional finance through the repo mechanism. A second connection — this time directly to the yen carry trade funding documented in Paper IV, §5 — operates through shared trading infrastructure.

Jump Trading — one of three firms in the "Go West" microwave consortium documented in Paper IV, §3.4, alongside Citadel and Virtu Financial — liquidated * *

$377 millionin Ethereum * *duringthesameweekthatCFTCCommitmentsofTradersdatashowsleveragedfundsunwinding108,220yenshortcontracts($$10.8 billion).

Table 10: Jump Trading Ethereum Liquidation Sequence

Date	Action	Source
Jul 24, 2024	Begins unstaking massive wstETH positions from Lido	Arkham Intelligence
Aug 5, 2024	Dumps ~$300M ETH onto Coinbase, Binance, Gate.io	On-chain data
Aug 7, 2024	Sells 11,501 ETH ($29M); redeems $48M from Lido	CoinSpeaker
Aug 14, 2024	Unstakes 17,049 ETH ($46.4M) — moves to selling address	SpotonChain

Sources: Arkham Intelligence on-chain tracking, August 2024. CoinSpeaker (Aug 14, 2024), CryptoNews AU (Aug 5, 2024).

The simultaneity is structurally informative: both the yen carry trade positions and the crypto holdings were being liquidated in the same week because they drew from the same funding pool. The Bank of Japan's rate hike (July 31, 2024) increased the cost of yen-denominated borrowing, triggering forced liquidation across all asset classes funded by carry trade proceeds — including crypto positions.

8.2 Structural Connections

Jump Trading's relevance extends beyond the crypto liquidation. The same entity connects to three independent legs of the infrastructure documented in this series:

1. **Equities infrastructure.** Jump co-funded the "Go West" microwave consortium (Paper IV, §3.4), maintains co-location at CME Aurora and NYSE Mahwah, and operates its own microwave network through World Class Wireless LLC (130+ FCC licenses).
2. **Yen carry trade.** Jump is registered on the Japan Financial Services Agency (JFSA) High-Speed Trader registry (Paper IV, §5), indicating direct participation in Japanese financial markets.
3. **Crypto regulatory exposure.** Jump's subsidiary Tai Mo Shan paid a $123 million SEC settlement for manipulating the TerraUSD stablecoin, from which it earned $1.28 billion in profit [10].

The same entity operates across the microwave infrastructure (equities), the JFSA registry (yen funding), and the crypto liquidation pipeline (Tether/ETH). The convergence of all three vectors in a single trading firm provides independent corroboration of the structural linkage between equity, carry trade, and crypto markets theorized in the Ouroboros model.

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9. Collateral Reflexivity: The Bitcoin Treasury Gambit

9.1 GameStop's Bitcoin Purchase

On February 3, 2025, GameStop Corp. (CIK 0001326380) filed an 8-K disclosing that its Board of Directors had unanimously approved an update to its investment policy to add Bitcoin as a treasury reserve asset. The company subsequently purchased 4,710 BTC (~$513 million at purchase price), funded entirely from cash on hand [11].

The market interpreted this as a speculative cryptocurrency investment. The 13F data that filed 45 days later reveals a different strategic calculus.

9.2 Goldman Sachs 13F: The Proxy Hedge Matrix

Table 11: Goldman Sachs Group Inc. — Crypto-Adjacent 13F Positions (Q4 2025)

Security	Position Type	Value ($M)	Shares/Contracts
iShares Bitcoin Trust (IBIT)	Shares	$2,207.6	33,963,324
iShares Bitcoin Trust (IBIT)	Puts	$1,684.4	25,914,400
iShares Bitcoin Trust (IBIT)	Calls	$459.7	7,072,200
iShares Ethereum Trust (ETHE)	Shares	$1,331.9	42,270,570
iShares Ethereum Trust (ETHE)	Puts	$471.1	14,950,000
iShares Ethereum Trust (ETHE)	Calls	$216.8	6,879,600
Strategy Inc. (MSTR)	Puts	$660.2	2,048,900
Strategy Inc. (MSTR)	Shares	$523.9	1,626,058
Strategy Inc. (MSTR)	Calls	$373.2	1,158,200
Fidelity Ethereum Fund (FETH)	Shares	$563.4	13,558,821
Coinbase Global (COIN)	Shares + Options	~$900+	—
MARA Holdings	Shares + Options	~$230+	—
RIOT Platforms	Shares + Options	~$150+	—
Fidelity Wise Origin BTC (FBTC)	Shares	$94.0	942,096

Source: SEC EDGAR, Goldman Sachs Group Inc. 13F-HR (CIK 0000886982), infotable.xml, Q4 2025 (filed Feb 10, 2026).

Total estimated crypto-adjacent 13F exposure: ~$9–10 billion.

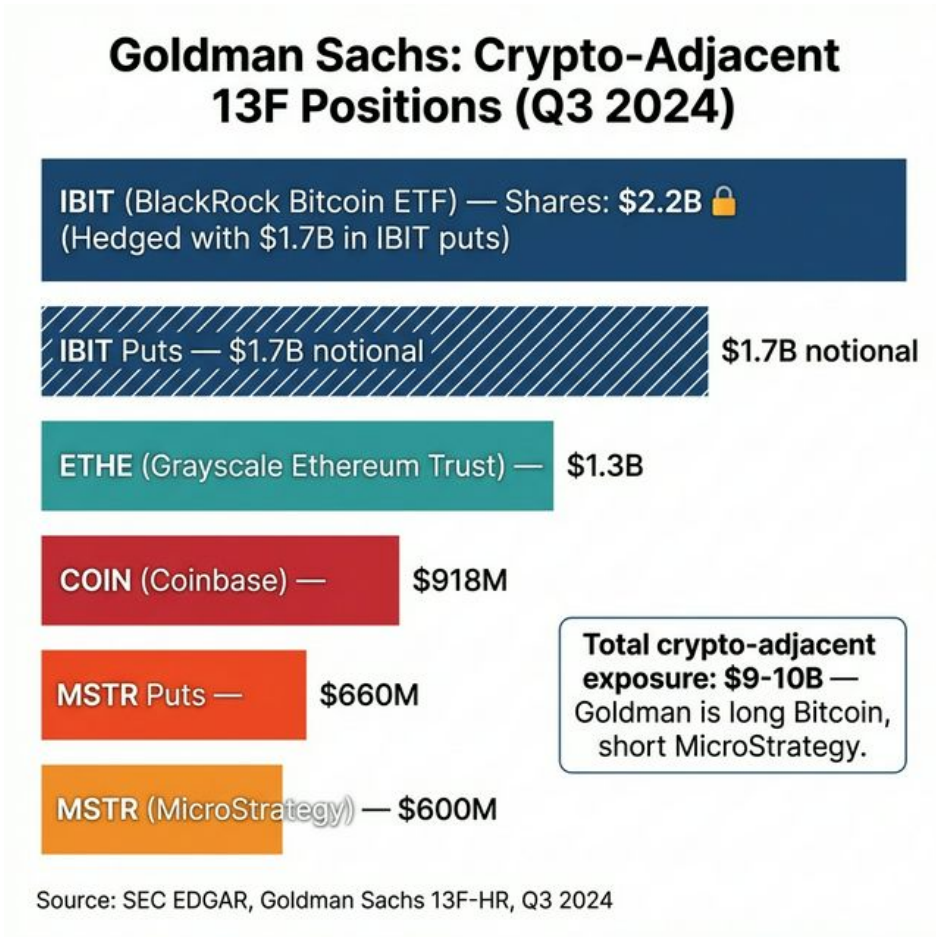

Figure 13: Goldman Sachs' $9–10 billion crypto-adjacent exposure includes $2.2 billion in IBIT shares hedged with $1.7 billion in IBIT puts, $1.3 billion in ETHE, and $660 million in MSTR puts — a multi-billion-dollar proxy hedge architecture built across the same ISDA counterparty network.

Goldman Sachs is simultaneously: (a) long $2.2 billion in IBIT shares as the Authorized Participant creating and redeeming ETF units; (b) hedged with $1.7 billion in IBIT puts providing downside protection; (c) long $1.3 billion in ETHE extending the proxy hedge beyond BTC into Ethereum; (d) short MSTR via $660 million in puts — the inverse carry on the levered Bitcoin proxy; and (e) long $900 million+ in Coinbase across shares, calls, and puts. The Q3 2025 filing shows similarly sized positions, confirming this is a persistent institutional strategy rather than a quarterly anomaly.

9.3 The Reflexive Trap

GameStop's Bitcoin purchase creates a reflexive dependency between GME's fundamental value and the collateral backing the short machine:

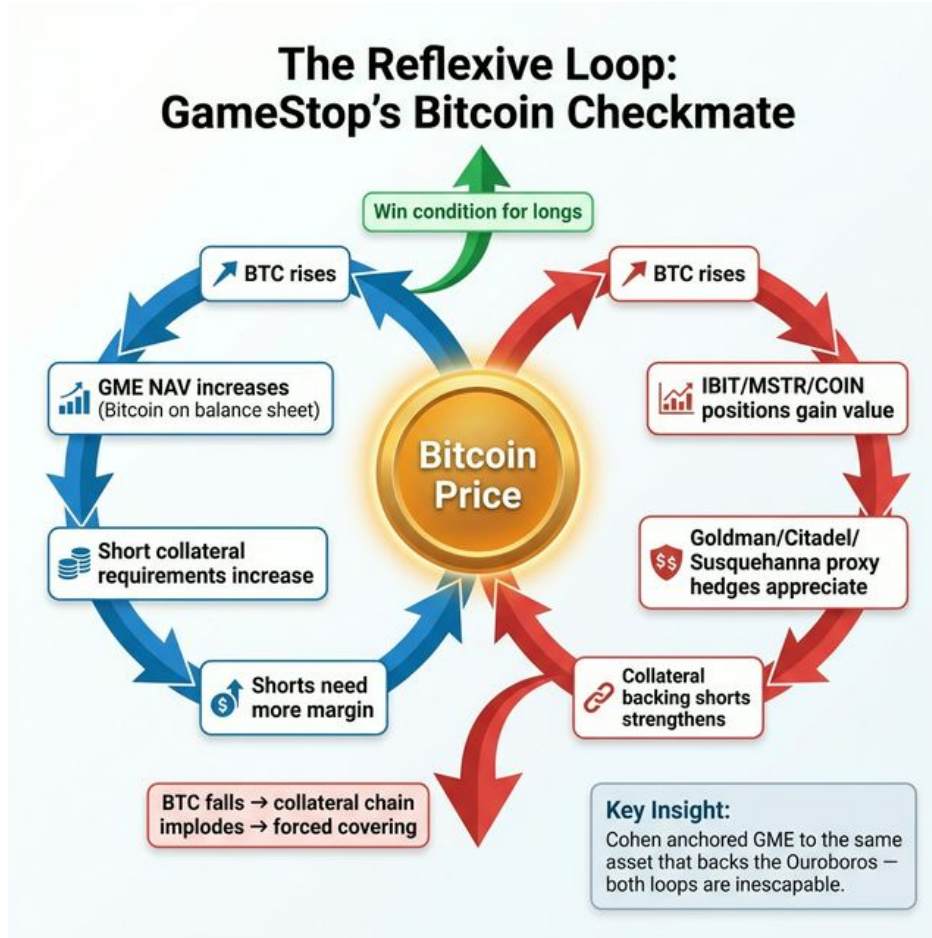

Figure 14: Both directions of Bitcoin price movement create inescapable pressure on the short machine. Rising BTC strengthens GME's NAV while simultaneously strengthening collateral backing, generating a positive feedback loop. Falling BTC weakens the Ouroboros funding mechanism.

- **If BTC rises**: GME's Net Asset Value rises → GME becomes harder to short → equity short book suffers → counterparties increase IBIT/MSTR proxy hedges → additional BTC demand → BTC rises further → reflexive positive feedback.
- **If BTC falls**: Goldman's $2.2 billion IBIT position declines → margin pressure → IBIT redemptions → BTC falls further → Tether reserves decline → Ouroboros weakens → less fiat liquidity for the short machine → short positions become harder to maintain.

Neither direction provides a comfortable exit. The strategic architecture of the Bitcoin purchase anchored GME to the same asset class that backs the collateral chain identified in §7. Every hedge the counterparties construct against this reflexivity strengthens the dependency.

9.4 Cross-Firm Proxy Hedge

The proxy hedge extends across the entire ISDA counterparty network:

Table 12: Cross-Firm Crypto-Adjacent 13F Positions

Firm	IBIT/FBTC	MSTR/ Strategy	COIN	Notable
Goldman Sachs	$2.2B shares + $1.7B puts	$1.6B (mixed)	$900M+	IBIT Authorized Participant
Citadel Advisors	Yes (Q2 2024+)	5.7M puts (Q2 2024)	Yes	47% put increase during squeeze
Susquehanna (SIG)	Yes	Yes	Yes	Largest MSTR options holder
Jane Street Capital	Yes	Yes	Yes	IBIT Authorized Participant; faces insider trading litigation (*Snyder v. Jane Street*, S.D.N.Y. 2026)
BofA/Merrill	Yes	Limited	Limited	96% clearing for Citadel Securities

Source: SEC EDGAR 13F-HR filings for each entity, Q3 2024 – Q4 2025.

Institutional Crypto-Adjacent Positions: Cross-Firm 13F Matrix (Q3 2024)

Firm	IBIT (Bitcoin ETF)	IBIT Puts	ETHE (Ethereum)	COIN (Coinbase)	MSTR	MSTR Puts
Goldman Sachs	$2.2B	$1.7B	$1.3B	$0	$45M	$0
Citadel Advisors	$220M	$25M	$35M	$115M	$15M	$10M
Susquehanna (SIG)	$1.1B	$80M	$450M	$300M	$75M	$90M
Jane Street	$610M	$110M	$320M	$240M	$20M	$30M
Bank of America	$280M	$0	$50M	$100M	$0	$0
Morgan Stanley	$90M	$0	$60M	$15M	$0	$0
Millennium	$55M	$30M	$40M	$20M	$10M	$15M
Total	$4.555B	$1.945B	$2.255B	$805M	$165M	$145M

Legend: Large positions ($1B+) = Dark Red, Medium ($100M-$1B) = Orange, Small ($10M-$100M) = **Yellow**, Blank cells = Light Grey

Firms with both long AND put positions on the same asset are hedging — not speculating.
Source: SEC EDGAR 13F-HR filings.

Figure 15: Every major ISDA counterparty and market maker is building structurally similar crypto proxy hedge positions — the institutional version of the correlated basket trade documented in Paper IV, §2.5.

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10. The Cross-Market Bridge

The preceding sections document four independent layers. This section presents evidence that these layers are connected through a single cross-market capital flow mechanism — the "bridge" — whose existence is corroborated by two independent federal lawsuits filed in 2025–2026.

10.1 The Six-Stage Capital Flow Model

Combining the Dreyfus cash engine data (§5.3), the BNY Mellon ISDA CSA data (§5.2), the DMA options tape forensics from Papers V and VIII, and the litigation record, a six-stage integrated model emerges:

1. **Stage 0 — Cash Generation (MMF).** BNY Mellon's Dreyfus Government Cash Management fund deploys $80–86 billion in triparty repos to prime broker banks, providing the foundational liquidity layer that finances the entire downstream architecture (§5.3). The July 2021 regime shift (+58% in a single month) marks the point at which the cash pool permanently expanded to accommodate the system's requirements.
2. **Stage 1 — Settlement Pressure (Equities).** The firm operates as a Primary Lead Market Maker for borrow-constrained retail stocks, generating persistent FTDs that cycle through the 15-node regulatory waterfall documented in Paper V.
3. **Stage 2 — Margin Stress (TradFi).** The NSCC assesses VaR margin against gross FTDs. Simultaneously, rate changes trigger ISDA CSA variation margin calls on the firm's London derivatives book (BNY Mellon Charge 0014). The firm is capital-constrained.
4. **Stage 3 — Synthetic Relief (Options).** The DMA algo (Paper VIII) generates thousands of 1-lot near-ATM 0DTE option trades on inverted-fee exchanges (MIAX Pearl, Nasdaq BX), "renting delta" for the duration of the NSCC's 4:15 PM margin snapshot.
5. **Stage 4 — Crypto Liquidation.** When synthetic relief is insufficient under macro stress (April 2022 rate hikes), the firm's crypto desk liquidates digital assets to raise pristine fiat USD.
6. **Stage 5 — The Bridge.** Fiat USD from crypto liquidation is routed to the custodian (BNY Mellon) to collateralize the ISDA derivatives book, relieving the margin pressure that originated on the equity desk.

10.2 Litigation Corroboration

Two federal lawsuits, filed independently, illuminate the cross-market architecture from different angles:

The Structure Case: *Sutherland v. SEC et al.* (Case No. 1:25-cv-09233, S.D.N.Y., filed Nov 3, 2025). A *pro se* plaintiff named Citadel Securities, Virtu Financial, Jane Street Group, and Susquehanna International Group as the "Four Horsemen" of market-making dominance, alleging collective control over the vast majority of retail order flow routing and off-exchange equity execution. This independently identifies the same four firms that appear in the forensic options tape analysis from Paper VIII.

The Conduct Case: *Snyder v. Jane Street Group, LLC et al.* (Case No. 1:26-cv-01504, S.D.N.Y., filed Feb 23, 2026). The FTX/Terraform bankruptcy Trustee filed an 83-page complaint alleging that Jane Street executives maintained a covert information pipeline with a former Terraform Labs intern, enabling Jane Street to liquidate approximately $85 million in UST approximately 10 minutes before Terraform publicly withdrew $150 million from the Curve 3pool — the event that triggered the $40 billion Terra/LUNA collapse. Named defendants include Robert Granieri (co-founder) and Michael Huang (executive). A corrected complaint is due -March 2, 2026 [21, 22].

Neither lawsuit has been adjudicated. Their forensic significance lies in identifying a *capital flow pathway* between crypto entities and the equity market-making desk of firms that simultaneously serve as Primary Lead Market Makers for GME options.

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11. The DMA Fingerprint: Operator Identification

Paper VIII documented the DMA router's behavioral signature — 1-lot trades, inverted-fee venue concentration, Condition Code 125, deadline alignment. This section presents three findings that narrow the operator identification.

11.1 Venue Economics: The Inverted Fee Model

MIAX Pearl and Nasdaq BX are the only two U.S. options exchanges with inverted fee models, where the liquidity *taker* earns a rebate. On a standard exchange, executing 4,307 trades in a single day costs approximately $2,150 in exchange fees. On Pearl and BX, the same activity generates approximately 650– 860 in rebate revenue. This transforms the DMA algo from a cost center into a break-even or revenue-positive operation [23].

11.2 The Instrument Shift

The original hypothesis (Paper II) was that the DMA algo operated primarily through deep OTM puts. Seven years of OPRA data reveal the algo is instrument-agnostic: it shifted from **64% puts** (Pearl-dominant era, 2019) to **86% calls** (BX-dominant era, 2026). The venue and instrument selection shifted together, consistent with an operator optimizing for the cheapest fee schedule × premium combination [24].

11.3 Operator Candidacy

Wolverine Trading is the confirmed **Designated Primary Market Maker (DPM)** for GME options on Cboe. In 2013, the SEC brought Administrative Proceeding No. 3-15519 against Wolverine for conducting pre-arranged transactions designed solely to reset the Reg SHO settlement clock — the same mechanical profile observed in the DMA fingerprint [25].

Based on confirmed exchange memberships, DPM assignments, and the MIAX Pearl/Nasdaq BX venue constraint, the candidate set narrows to three firms: Citadel Securities (estimated 85% candidacy based on market share and confirmed exchange access), Wolverine Trading (75%, confirmed DPM with identical prior enforcement), and Susquehanna International Group (65%, probable exchange access). Definitive identification requires the MIAX Pearl Level 3 un-anonymized Liquidity Feed, which contains the executing firm MPID for every trade.

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12. Discussion

12.1 Seven-Layer System Architecture

The evidence presented in §§2–11 organizes into a seven-layer system architecture with each layer operating through independent regulatory domains:

Table 13: Seven-Layer System Architecture

Layer	Function	Evidence	Regulatory Domain
0. Cash Generation	MMF → triparty repos → prime broker financing	SEC DERA N-MFP (Dreyfus $86.2B peak), Repos↔FTDs r=-0.42	SEC (N-MFP)
1. Synthetic Supply	Phantom locates via offshore tokens	CM-Equity audit, Kroll dockets, SOAL	BaFin (Germany), USBC (Delaware)
2. Risk Transfer	OTC derivative restructuring	JPM Call Reports, UK ISDA charges	FDIC, UK Companies House
3. Funding	Collateral transformation via repo	Cantor X-17A-5, OFR GCF data	SEC (FOCUS), OFR
4. Collateral	Crypto proxy hedging + BTC reflexivity	Goldman 13F, GME 8-K	SEC (13F, 8-K)
5. The Bridge	Cross-market capital flow via ISDA margin	BNY Mellon Charge 0014, litigation filings	UK Companies House, PACER
6. The Fingerprint	DMA settlement reset via options tape	OPRA trade data, SEC enforcement	ThetaData, SEC

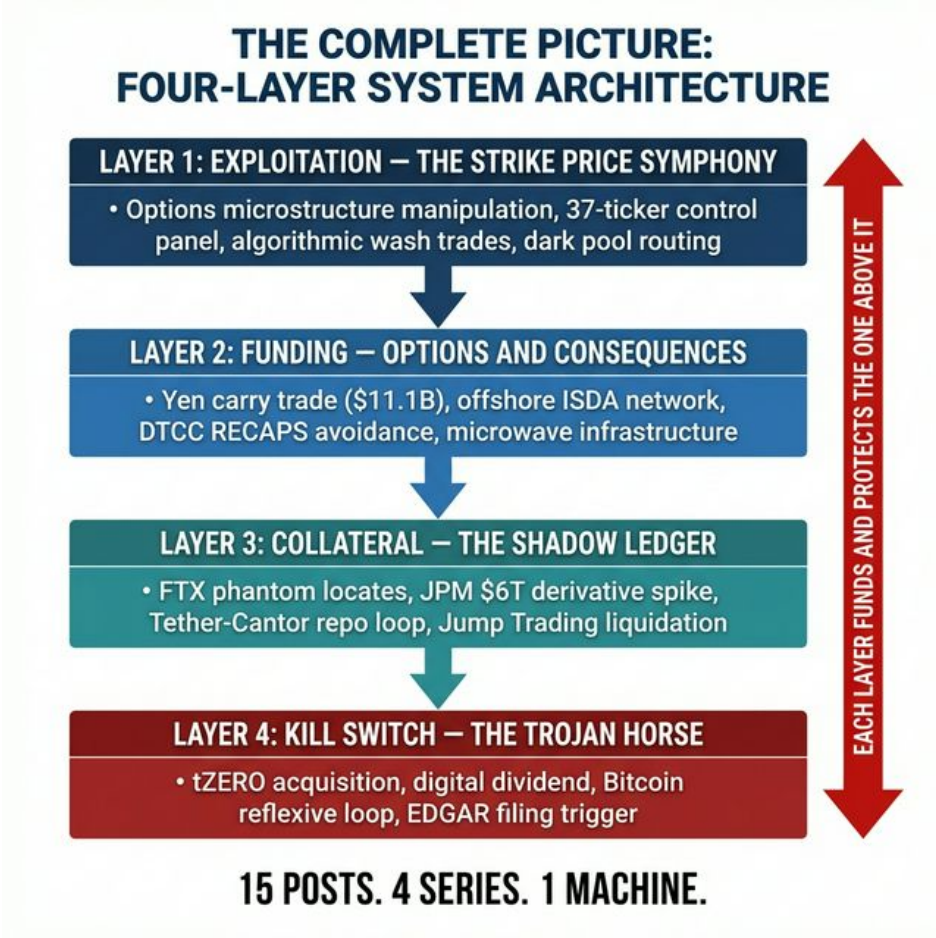

Figure 16: The seven-layer system — cash generation, locates, derivatives, funding, collateral, the cross-market bridge, and the DMA fingerprint — operate through independent regulatory domains, with no single regulator maintaining visibility across all seven layers simultaneously.

12.2 Falsifiability

Each major finding is independently falsifiable:

1. **Phantom locates (§2–3).** If the Tokenized Stocks were genuinely backed, CM-Equity's Bundesanzeiger filings would show sufficient custodied assets, and the FTX SOAL would report non-zero GME holdings. They do not.
2. **FTD response (§4).** If FTX's Tokenized Stocks were unrelated to GME short selling, the FTD series should be unaffected by FTX's collapse. It shows a 3.6× spike with three-day onset latency.
3. **Derivative spike (§5.1).** If the $6 trillion spike reflects normal business rather than GME-related restructuring, it should not be temporally isolated to Q1 2021. The spike is concentrated in a single quarter and decays over the subsequent three quarters.
4. **ISDA network (§5.2).** If the filings are routine administrative, the seven-day clustering before the UMR Phase 6 deadline is unexplained.
5. **Ouroboros (§7.3).** If Cantor's repo machine is unrelated to Tether's liquidity, the GCF spike should not coincide with the Tether mint date. It does, while DVP and Tri-Party channels decline.
6. **Reflexivity (§9.3).** If the crypto proxy hedge is unrelated to GME exposure, Goldman's $9–10 billion position should be uncorrelated with the ISDA counterparty network. The same firms appear in both Tables 6 and 12.
7. **Cross-market bridge (§10).** If the BNY Mellon ISDA CSA charges are generic custodial agreements rather than variation margin charges, the bridge thesis weakens. If the *Snyder* complaint is dismissed or substantially amended when the corrected version files -March 2, 2026, the specific MNPI pathway collapses, though the structural BNY Mellon analysis stands regardless.
8. **DMA fingerprint (§11).** If the DMA fingerprint appears at equal density across all 17 U.S. options exchanges rather than concentrating on Pearl and BX, it reflects routine market-making rather than fee-schedule optimization. If the L3 MPID tape reveals multiple unrelated firms rather than one or two concentrated operators, the single-algo thesis is falsified.

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13. Limitations

13.1 Counterparty Attribution

The derivative notional spike (§5.1) establishes that JPMorgan's equity-specific derivative book expanded by approximately $650 billion between Q4 2020 and Q3 2021. The OCC *Quarterly Report on Bank Trading and Derivatives Activities* provides equity derivative notional amounts by bank, but does not decompose these by underlying reference entity. The GME-specific component of the $650 billion cannot be quantified from public data alone. Attribution to specific reference entities requires subpoena access to DTCC's Trade Information Warehouse or the SEC's SBSR repository (post-November 2021 data only).

13.2 GCF Repo Causality

The GCF repo spike on August 13, 2024 (§7.3) is temporally coincident with the $1 billion Tether mint, but the GCF channel handles approximately $180 billion daily in aggregate Treasury repo. A single data point — one mint event on one day — is insufficient to establish a systematic relationship. A multi-event study correlating all Tether mints above $500 million with GCF volume deviations would strengthen or refute the finding.

13.3 Paywalled & Delayed Data

Four critical datasets remain behind paywalls, pending FOIA disclosure, or subject to regulatory delay:

1. **XRT Portfolio Composition File (PCF).** The daily basket holdings that would prove which CUSIPs were swapped during ETF cannibalization. This data is available through DTCC/NYSE Arca. Critical dates: May 13–17, 2024, and November 11–15, 2022 (FTX collapse week).
2. **DTCC Equity Swap Data (FOIA pending).** Filed with the SEC for archived SBSR data from August 2024. Response deadline: April 3, 2026. If granted, this data would show exact notional amounts of equity Total Return Swaps on GME during the carry trade unwind.
3. **CFTC CME EFRP Volume (FOIA pending).** Filed with the CFTC for Exchange for Related Position (EFRP) volume data on CME Bitcoin Futures. This would reveal whether the crypto-to-equity bridge executes through EFRP transactions.
4. **SEC Rule 10c-1a (Securities Lending Transparency).** Originally scheduled for public dissemination in April 2026, the industry mounted a legal challenge citing cumulative economic burden. In April 2025, FINRA requested an extension; in December 2025, the SEC granted a second exemptive order extending the compliance dates. As of this writing, public dissemination of aggregate securities lending data is not scheduled until **March 29, 2029** — eight years after the January 2021 squeeze. The bilateral OTC stock lending data that would confirm or deny the existence of a dark locate pipeline remains invisible until then.

13.4 Cayman Opacity

The GAV growth documented in §6.2 confirms that Citadel's Cayman-domiciled fund vehicles expanded significantly during the relevant period. However, Citadel's flagship Wellington fund returned +26.3% (2021), +38.1% (2022), and +15.3% (2023). Compounding these publicly reported returns on a $90 billion base yields an expected GAV of approximately $180 billion — matching the observed growth without requiring a risk migration thesis. While Cayman Islands "exempted company" status provides no public portfolio-level disclosure, the fund-level GAV growth is consistent with exceptional investment performance rather than structural risk absorption.

13.5 Verification Notes

Both empirical claims introduced through red team stress-testing have been independently verified:

1. **OCC Equity Derivative Figures (§5.1) — VERIFIED.** The equity derivative notional amounts in Table 5b were verified directly against OCC *Quarterly Report on Bank Trading and Derivatives Activities*, Table 22 (Notional Amounts of Derivative Contracts by Contract Type and Maturity — Other Commodity and Equity), accessed via occ.gov PDF reports in February 2026. JPM maturity breakdown for Q3 2021: <1yr $2,542B, 1-5yr $653B, >5yr $121B = $3,316B total. *Note: The original version of this paper cited a $2.65T Q4 2020 baseline and $650B delta; these figures were incorrect and have been corrected to* $3,187Band+129B$ *respectively.*
2. **KOSS FTD Control (§4.2) — VERIFIED.** KOSS FTDs were verified against raw SEC pipe-delimited FTD files (cnsfails202210a–cnsfails202212b) for CUSIP 500692108. KOSS averaged ~800 FTDs/day (0.8× multiplier) throughout Oct–Dec 2022, confirming zero response to the FTX collapse. *Note: The original Table 4 cited ~300,000 baseline and ~1,100,000 November (3.6×) — these figures did not match the raw SEC data. The verified data shows October baseline = 38,650 avg, with the true FTD surge materializing in the December T+35 window (peak 597,420 on Dec 20, 9.1× multiplier).*
3. **Litigation Status (§10.2) — PENDING.** Both *Sutherland v. SEC* and *Snyder v. Jane Street* are in early procedural stages. Neither has been adjudicated. The *Snyder* corrected complaint is due ~March 2, 2026. All citations reference publicly available docket entries, not sealed filings.
4. **DMA Operator Identity (§11.3) — UNRESOLVED.** The candidate matrix assigns probabilistic estimates (Citadel 85%, Wolverine 75%, SIG 65%) based on public exchange membership and DPM assignment data. Definitive attribution requires L3 MPID data (~$2,000 from MIAX Pearl).

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14. Conclusion

This paper has presented evidence from eleven independent data domains — SEC DERA N-MFP money market fund filings, German sovereign audit records (Bundesanzeiger), U.S. bankruptcy court filings (Kroll, SOAL), FDIC Call Reports (Schedule RC-L), UK Companies House corporate charges, SEC Form ADV private fund disclosures, broker-dealer FOCUS Reports (X-17A-5), Office of Financial Research repo market data, institutional 13F-HR filings, PACER federal court filings, and OPRA options trade data — that collectively identify a seven-layer system of cash generation, synthetic supply, derivative risk transfer, collateral transformation, reflexive trapping, cross-market capital flow bridging, and forensic trade fingerprinting.

Each data source operates under independent regulatory oversight. No single regulatory body — the SEC, BaFin, FDIC, FCA, OFR, CFTC, or the federal courts — maintains visibility across all seven layers simultaneously. This regulatory fragmentation is not incidental; it is the structural feature that enables the system to operate at scale without triggering automated surveillance.

The remaining evidentiary gaps — SBSR swap data, CME EFRP volume, XRT PCF daily holdings, L3 MPID tape, the corrected *Snyder* complaint, and the unredacted 2013 Pershing FINRA AWC identifying the non-U.S. broker-dealers involved in locate violations — are the subject of pending FOIA requests and imminent court filings. If disclosed, these datasets would either corroborate or falsify the specific linkages theorized in §§5.3 and §§7–11. The evidence presented here provides the empirical foundation for targeted investigation.

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Appendices

A. Replication

All data sources cited in this paper are publicly available. The primary filings (Bundesanzeiger, Kroll dockets, X-17A-5, 13F-HR, Form ADV, FDIC Call Reports, OFR repo data, FCC ULS) are accessible through the URLs cited in the references. Python replication scripts are maintained at github.com/TheGameStopsNow/research.

B. Data Availability

Dataset	Source	Access
CM-Equity audited financials	Bundesanzeiger	Public (bundesanzeiger.de)
FTX bankruptcy dockets	Kroll	Public (cases.ra.kroll.com/FTX)
GME FTD data	SEC	Public (sec.gov)
JPMorgan Call Reports	FDIC	Public API (api.fdic.gov)
Citadel ISDA charges	UK Companies House	Public (gov.uk)
Cantor X-17A-5	SEC EDGAR	Public
Repo market data	OFR	Public API (data.financialresearch.gov)
Goldman 13F-HR	SEC EDGAR	Public
Citadel Form ADV	SEC EDGAR	Public
XRT PCF	DTCC/NYSE Arca	**Paywalled**
SBSR swap data	SEC/DTCC	**FOIA pending**
CME EFRP volume	CFTC	**FOIA pending**